



**Politecnico di Milano**

# **Dipartimento di Elettronica, Informazione e Bioingegneria**

**prof. Fabrizio Ferrandi**

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## **Parallel Computing– 1<sup>st</sup> mid-term test– Tuesday, November 9<sup>th</sup> 2023**

**Polimi ID** \_\_\_\_\_

**Surname** \_\_\_\_\_ **Name** \_\_\_\_\_

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:30m.

**Exercise 1 (4 points)** \_\_\_\_\_

**Exercise 2 (4 points)** \_\_\_\_\_

**Exercise 3 (4 points)** \_\_\_\_\_

**Exercise 3 (4 points)** \_\_\_\_\_

## Exercise n. 1

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid)

- A. Please list the characteristics of the PRAM model that simplify the time complexity estimations. (1)

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- B. Is the time required to access the memory different from the time taken to access the main memory in a PRAM model? (1)

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- C. List the key points of Gustafson's law. (2)

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## Exercise n. 2

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid).

A. What does it mean conditional execution in a data-parallel program? (1)

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B. Is prefetching reducing stalls? What does it mean prefetching/stall in modern CPUs?(2)

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C. Please explain the differences between SIMD against multi-core parallel execution. (1)

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### Exercise n. 3

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid).

A. Please describe how multi-threading can be implemented at the core level. (1)

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B. Please explain why the memory bandwidth could be a critical resource. (1)

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C. Please explain how the CUDA threads are executed. (2)

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#### Exercise n. 4

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Is it tiling related to the GPUs' shared memory? Please explain why and when. (1) **This question can be skipped in case of a passed challenge.**

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Please define what memory coherency means. (1)

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- B. Which operations ordering are considered by a sequentially consistent memory system? (1)

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- C. Why custom computing and heterogeneous systems are relevant today? (1)

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